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Case report

Acute pancreatitis following COVID-19 vaccine: A case report and brief literature review

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ABSTRACT

Vaccination is the most effective way to overcome COVID-19 morbidity and mortality. However, Covid-19 vaccines may cause potential adverse effects. We reported a 28-year-old healthy woman who was referred to the emergency department with a chief complaint of severe abdominal pain, nausea and hemoptysis. She has received two doses of COVID-19 vaccine (Sinopharm BIBP). Similar this time, three days after the injection of the second dose of the Sinopharm BIBP COVID-19 vaccine, abdominal and flank pain appeared, for which she has referred to the emergency department. After necessary tests and pancreatitis was confirmed, we started fluid therapy, plasmapheresis, gemfibrozil and insulin for patient management.

The COVID-19 vaccines may lead to acute pancreatitis. The mechanism of pancreatitis caused by COVID-19 vaccines is unclear. Acute pancreatitis can develop after COVID-19 vaccination. This process can even happen a few months later. Therefore, to better diagnosis and prevention of long-term complications, it is necessary to measuring the lipase or amylase in patients that received COVID-19 vaccine if abdominal pain was occurred.

1. Introduction

In late 2019, a new coronavirus was first identified as causing pneumonia in China. In February 2020, coronavirus Disease 2019 (COVID-19) had been officially announced by the World Health Organization (WHO) [1].

Every individual is at risk for infection with the COVID-19 virus [2]. As of March 2022, more than six million deaths from COVID-19 have been reported worldwide. Even though COVID-19, is known as a respiratory disease, this disease can cause other problems such as cardiac damage, renal, skin, neurological, gastrointestinal, and hematologic problems, etc., and indirectly lead to death [3,4].

Since the spread of COVID-19, the greatest hope for prevention and return to normal life has been the development of an effective vaccine. We currently have several vaccines available to prevent this disease. RNA or DNA-based vaccines, viral vectors, inactivated viruses, and protein subunits, are the four main types of COVID-19 vaccines [5]. Sinopharm BIBP, CoronaVac (Sinovac COVID-19 vaccine), COVIran Barekat, and Covaxin (BBV152) are inactivated viral vaccines that consist of viral particles grown in a culture

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medium and then inactivated to lose the capacity to produce the disease while stimulating the immune response [6,7]. The efficacy and safety of these vaccines have been proven in various clinical studies and have been approved by monitoring institutions such as Food and Drug Administration (FDA) and WHO [8]. Despite the unique and important role of COVID-19 vaccines in preventing mortality and reducing treatment costs, it is nevertheless important to consider the potential adverse events that these vaccines may cause [9].

With the increase in the number of vaccine injections in different countries and the increase in the vaccinated population, more side effects can be reported in this area. Mild side effects such as injecting site pain, headache, pain in the extremities, fatigue, fever, etc. have been reported with injections of COVID-19 vaccines [10]. In addition, severe and important complications such as unilateral optic neuritis [11], autoimmune encephalitis [12,13], Stevens-Johnson syndrome (SJS) [14], Thrombotic Thrombocytopenic Purpura [15], cerebral venous thrombosis [16] have been reported with the injection of inactivated COVID-19 vaccine (especially with Sinopharm BIBP and Sinovac).

The purpose of this patient report is to emphasize the role of the clinical staff in evaluating the long-term adverse events of new vaccines. This information may be necessary to make physicians aware of timely intervention and to establish guidelines for the protection of persons at risk.

2. Cases presentation

This study complied with the Declaration of Helsinki's ethical standards [17]. It was presented and numbered in the Ethics Committee of Mazandaran University of Medical Sciences with the ethics number: IR. MAZUMS.REC.1401.16151. The patient was a 28-year-old healthy woman who was referred to the emergency department on February 15, 2022, with a chief complaint of severe abdominal pain. Abdominal pain started abruptly at noon on the day of admission. The nature of the pain is continuous, and it aggravates by bending forward and radiating to the sides. She had several times of nausea and hemoptysis and she was afflicted with aphagia and was unable to eat. She has not had defecation for about four days.

Her vital signs immediately after entering the emergency department were as follow: blood pressure: 130/70 mmHg, temperature: 37 °C, heart rate: 101 beats/min, and respiratory rate: 13 breaths per minute. On physical examination, epigastric tenderness and pain in the right upper quadrant (RUQ) area were observed. Height, weight, and BMI (body mass index) were 172 cm, 76 kg, and 25.7 respectively.

In the history of the patient's medical problems and hereditary diseases, she had no diagnosed underlying or hereditary diseases, autoimmune disorders or connective tissue diseases. She mentioned only irregular menstrual periods. In her drug history, she only used a Cyproterone Compound tablet to regulate menstruation. She has no history of smoking, alcohol consumption or the use of illegal drugs. In addition, she did not mention any specific food or drug allergies.

Table 1
Initial laboratory data in the emergency department.

Lab data parameter	Reference Range	Result
WBC	4000 to 1000/mm3	8400/mm^3
PLT	145,000 to 450,000/mm3	196,000/mm3
Hb	12.3–15.3 g/dL	13 g/dL
Blood sugar	90–110 mg/dL	203 mg/dL
Urea	13–40 mg/dL	18 mg/dL
Creatinine	0.5–1.3 mg/dL	0.4 mg/dL
AST	5–40 U/L	44 U/L
ALT	5–40 U/L	80 U/L
ALP	64–306 U/L	119 U/L
Bil.T	0.2–1.2 mg/dL	0.7 mg/dL
Bil.D	0–0.3 mg/dL	0.32 mg/dL
K	3.5 to 5.5 mEq/L	3.6 mEq/L
Na	135 to 145 mEq/L	143 mEq/L
Calcium	8.5–10.5 mg/dL	7.6 mg/dL
Albumin	3.5–5.5 g/dL	4.4 g/dL
Mg	1.8–2.2 mg/dL	2.2 mg/dL
P	2.5–4.5 mg/dL	2.4 mg/dL
C.R.P	Less than 6 mg/L	25.6 mg/L
ESR	0–20 mm/hr	48 mm/h
TG	Less than 200 mg/dL	1562 mg/dL
Amylase	Less than 100U/L	1079 U/L
Lipase	Less than 60U/L	156 U/L
pH	7.35 to 7.45	7.30
HCO3	22–28 mmol/L	20.7 mmol/L
PCO2	35–45 mmHg	42 mmHg
Ck-MB	Less than 24 U/L	19 U/L

Abbreviations: White blood cells (WBC), Platelet count (PLT), Hemoglobin (Hb), Aspartate aminotransferase (AST), Alanine transaminase (ALT), Alkaline phosphatase (ALP), Total bilirubin (Bil.T), Direct bilirubin (Bil.D), Potassium (K), Sodium (Na), Magnesium (Mg), Phosphorus (P), C-Reactive protein (CRP), Erythrocyte sedimentation rate (ESR), Triglycerides (TG), power of hydrogen (pH), Creatine kinase-MB (CK-MB).

An important point in the taken history was related to the injection of the COVID-19 vaccine. The patient received two doses of the COVID-19 vaccine (Sinopharm BIBP), and the last vaccination was about 3 months ago (The first dose was in late September 2021 and the second dose was injected in early November 2021). Three days after the injection of the second dose of the Sinopharm COVID-19 vaccine, abdominal pain, and flank pain appeared, for which she was referred to the emergency department, and in the examinations, all the tests and imaging were normal and she was discharged with supportive treatments. About a week after the second dose of the vaccine, she suffered severe pain in her legs and then numbness in her left leg. During an outpatient visit to a neurology clinic, naproxen, gabapentin, vitamin B1, and magnesium due to neuropathy diagnosis were prescribed for her.

After taking a complete history, the patient was admitted with a possible diagnosis of acute pancreatitis. In the emergency department, she was first given an NPO regimen, and fluid therapy with normal saline (3 L) was started. Drug administration with Pantoprazole, Pethidine, and Meropenem was also started in the emergency department. Necessary tests such as CBC, BS, BUN, Cr, AST, ALT, ALP, Bil (T, D) Na, K, Ca, Mg, P, Alb, CRP, ESR, TG, Amylase, Lipase, ABG and Ck-MB were requested and the ultrasounds of the abdomen and pelvis were also performed.

In the ultrasound report, the liver was normal in echotexture and demonstrates no evidence of dilated intrahepatic ducts, and no gallstones were noted. Fluid in the sub-hepatic space, peri-splenic and paracolic gutter was observed. After about 2 hours of admission, the results of the initial laboratory tests were reported, and according to the laboratory results, and peri-pancreatic fluid in sonographic findings the diagnosis of pancreatitis was confirmed (Table 1).

Gallstones, trauma or pancreatic duct injury, alcohol, hypertriglyceridemia, genetic disorders, hypercalcemia, infections (viruses, bacteria, or fungi), and toxins, are major etiologies of acute pancreatitis [7]. Our patient had no known hereditary or autoimmune disorders. The history of trauma, alcohol consumption, and exposure to toxins was also negative. In addition, according to the results of ultrasound and laboratory findings, the possibility of gallstones and hypercalcemia were ruled out. Therefore, infection and hypertriglyceridemia were considered possible causes of pancreatitis in this patient. Generally, in patients with hypertriglyceridemia, primary and secondary disorders coexist. Nearly all patients with severe hypertriglyceridemia have both genetic and acquired

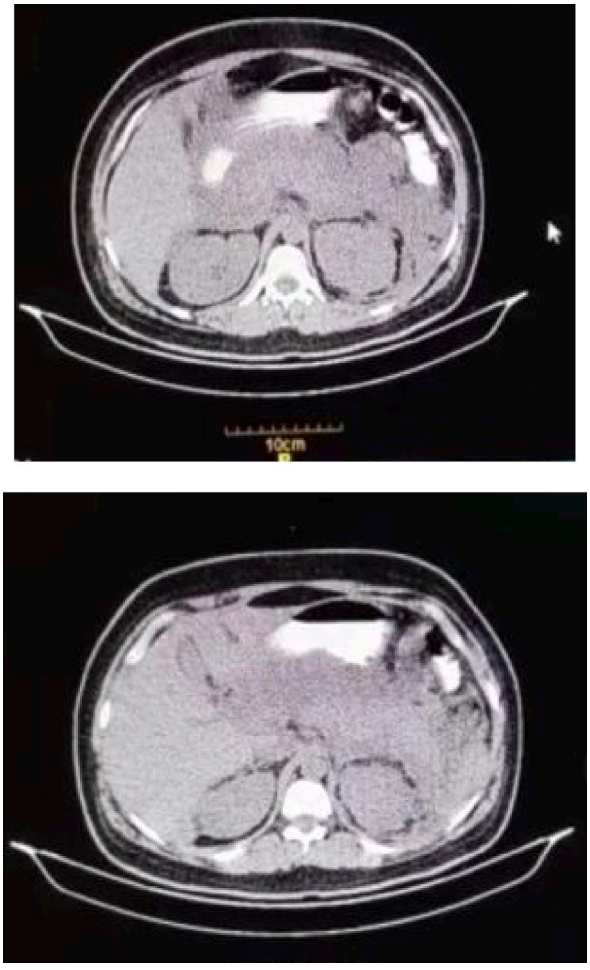


Fig. 1. Computed Tomography (CT) scan of the abdomen. Homogeneous enlargement of the pancreas, extensive peri-pancreatic fat and peri-pancreatic fluid was observed.

conditions. Pregnancy, obesity, diabetes mellitus type 2, renal disease, hepatocellular disease, and chronic inflammatory diseases can cause hypertriglyceridemia in at-risk patients. Our patient did not have a known and proven genetic disorder and did not have any of the above risk factors. Infections and viruses can lead to pancreatitis. Although this patient did not have a finding in favor of viral or bacterial infection, however, due to the use of the inactivated vaccine, and the similarity between the viral vaccine and the SARS-CoV-2 virus, the possibility of pancreatitis caused by the inactivated viral vaccine was considered. The stress of a severe inflammatory response to acute pancreatitis can cause an increase in serum triglycerides.

Then the patient was transferred to the gastrointestinal ward due to the high triglyceride in the initial tests. To manage the treatment of pancreatitis, insulin (regular) was started at a dose of 6 units per hour and a blood sugar level was performed every 6 hours. She was also given gemfibrozil 600 mg every 12 hours and omega-3 capsules, every 12 hours.

The patient complained of a tingling sensation in the face and fingers, and the Trousseau and Chvostek signs tests were positive. Calcium levels were requested urgently and according to the reported results; calcium therapy was started for the patient.

The patient, then, complained of respiratory distress and tachypnea, and was transferred to the intensive care unit (ICU) for better monitoring and continued treatment. Also, due to high triglyceride, consultation with the endocrine service and hematology-oncology service was requested for plasmapheresis. According to the hematology consultation, the patient underwent 3 L of plasmapheresis, and calcium levels were performed every 3 hours during plasmapheresis and then every 6 hours after its completion. According to endocrine counseling, regular insulin intake increased to 10 units per hour. Due to the patient's agitation, frequent restlessness and lack of cooperation, psychiatric counseling was also requested.

After plasmapheresis completion, the patient had a fever; and a complete Computed Tomography (CT) scan of the abdomen, pelvis, and chest, to assess for complications of pancreatitis, was requested (Fig. 1.).

According to the obtained graphs, evidence of pleural effusion and ascites was seen. Therefore, meropenem was started and an abdominal tap (paracentesis) and ultrasound-guided pleural biopsy were performed.

After these procedures, the patient had a fever again, and infectious consultation was requested. According to the infectious counseling, blood, urine, and secretion cultures were requested in terms of fungal infection and evaluation of lactate levels. Vancomycin was also added to the patient's antibiotic regimen.

Blood culture and secretion results, showed no trace of bacteremia or fungal infection and the patient's fever broke and the antibiotics continuation was stopped.

Finally, the patient was discharged to the gastroenterology clinic in good general condition based on an outpatient referral. Table 2 showed post-treatment laboratory data.

3. Discussion

The patient presented was a healthy 28-year-old woman without any risk factors or conditions for developing pancreatitis and had no family history of pancreatitis. A noteworthy point in our patient was the onset of early symptoms in the form of abdominal pain, three days after the injection of the second dose of the COVID-19 vaccine, followed by limb numbness and peripheral neuropathy. Then, about three months after the last dose of the vaccine, she was again afflicted with abdominal pain and was hospitalized with a diagnosis of acute pancreatitis.

Acute pancreatitis is defined as inflammation of the pancreas that results in damage to the pancreatic parenchyma [18]. Severe epigastric pain, and imaging findings are major criteria for acute pancreatitis diagnosis. Serum amylase elevation (greater than three times the upper limit of normal), lipase elevations, elevations in C-reactive protein (CRP), hypocalcemia, and other metabolic abnormalities are the most frequent laboratory findings in acute pancreatitis [19]. As demonstrated in Table 1 many aspects of the laboratory data of our patient were compatible with acute pancreatitis.

Acute pancreatitis can occur due to a variety of reasons, including hypertriglyceridemia, certain medications, toxins, gallstones, alcohol, and autoimmune infections [20]. In our patient, nearly all possible causes of pancreatitis were ruled out.

About half of patients admitted with a COVID-19 diagnosis have gastrointestinal symptoms in addition to respiratory symptoms [21]. Numerous cases of COVID-19-induced pancreatitis have been reported [22–24]. It shows that the SARS-CoV-2 virus can cause damage to the pancreas [25]. In some studies, damage to the pancreas has been observed in 15–20% of COVID-19 cases [26]; however, at present, the association between COVID-19 and acute pancreatitis is not based on any conclusive evidence.

The cause of COVID-19-induced pancreatitis is not fully understood; it appears that antibodies produced against the SARS-CoV-2

Table 2
Post-treatment laboratory data. The results demonstrate an improvement after treatment.

Lab data parameter	Reference Range	Result
Urea	13–40 mg/dL	9 mg/dL
Creatinine	0.5–1.3 mg/dL	0.8 mg/dL
K	3.5 to 5.5 mEq/L	4.5 mEq/L
Na	135 to 145 mEq/L	138 mEq/L
Calcium	8.5–10.5 mg/dL	8.6 mg/dL
Mg	1.8–2.2 mg/dL	2.3 mg/dL
TG	Less than 200 mg/dL	268 mg/dL
Amylase	Less than 100U/L	64 U/L

Abbreviations: Potassium (K), Sodium (Na), Magnesium (Mg), Triglycerides (TG).

spike and nucleoprotein can cross-react with many tissue antigens in the human body [27,28]. Nevertheless, the COVID-19 virus needs the angiotensin converting enzyme-2 (ACE2) receptor for binding to human cells. The amount of this receptor in the gastrointestinal tract is 100 times that of the respiratory tract [29]. The binding of the virus to the ACE2 receptor in the pancreas can lead to damage to the pancreas.

COVID-19 vaccines are the most effective ways to prevent this disease. Many different types of vaccines are produced in the world, such as mRNA-based or inactive and viral vectors. Since COVID-19 vaccines are new, it is reasonable to assume that we have not yet seen many of the possible adverse effects that may occur in patients who have received them. There have been some case reports that demonstrated the association between COVID-19 vaccines and the incidence of acute pancreatitis [30–32].

Table 3 demonstrated some clinical case reports and factors associated with COVID-19 vaccines induced acute pancreatitis. These reports have often occurred shortly after the BNT162b2 (Pfizer/BioNTech COVID-19 mRNA vaccine) [33–35]. As shown in Table 3, in eight patients, pancreatitis occurred after the first dose, in four patients after the second dose, and in one patient after the third dose of vaccination. One patient experienced pancreatitis after both the first and second doses. Except for one patient who reported chronic alcohol consumption, no direct risk factor for pancreatitis was observed in other patients. As mentioned, in all cases, pancreatitis was caused by the BNT162b2 vaccine. To our knowledge, we for the first time reported acute pancreatitis after the Sinopharm BIBP vaccine.

The mechanism of pancreatitis caused by COVID-19 vaccines is unclear. Similar to COVID-19 disease, the amino-acid similarity between the viral vaccine used and the body's antigens could lead to an autoimmune reaction. The inactivated virus is used in the Sinopharm BIBP vaccine platform. These inactive viruses can no longer reproduce, but their proteins, including the spike protein, remain intact. Similar to COVID-19, antibodies produced against the spike protein can lead to pancreatitis. Hence, in the pancreatic arteries, a high level of ACE2, is reported. Inflammatory cytokines can increase the incidence of ACE2 in the pancreas [44]. By binding to an inactivated virus (vaccine) containing the spike protein to pancreatic ACE2 receptors, there is a possibility of pancreatitis [28]. In other hand, COVID-19 vaccine can cause hypertriglyceridemia [45] and this secondary hypertriglyceridemia, can lead to pancreatitis.

Eventually, the presented patient, underline the possible adverse event of the COVID-19 vaccine and the importance of measuring lipase or amylase in patients that received COVID-19 vaccine if patients experience abdominal pain.

4. Conclusion

It is important to note that COVID-19 vaccines are still the most effective way to prevent and reduce mortality, and the incidence of adverse event should be weighed against the potential benefits of these vaccines. Acute pancreatitis can develop after COVID-19 vaccination. This process can even happen a few months later. Therefore, it is better to be more careful for choosing the type of vaccine for the high-risk groups.

Table 3
Case reports and clinical characteristics of patients with Covid-19 vaccine-induced acute pancreatitis.

Study (year)	Age/ Sex	Type of vaccine	Causative dose	The duration of the vaccine injection, until the onset of symptoms	Concomitant diseases or conditions	Ref
Rajib Kumar Dey (2022)	24/F	BNT162b2 [a]	First dose	1 week	Pregnancy	[36]
Ankoor H. Patel (2022)	63/M	mRNA-based [b]	Second dose	1 month	No known past medical history	[37]
Omar Fakhreddine (2022)	64/M	BNT162b2	Third dose	Few days [c]	No known past medical history	[38]
Alrashdi Mousa N (2022)	22/F	BNT162b2	First dose	1 week	No known past medical history	[39]
Brittain, Connor (2021)	46/M	BNT162b2	First and second dose	2 days after each dose	Hypertension	[40]
Riya Kaur Kalra (2022)	14/F	BNT162b2	Second dose	3 days	No known past medical history	[41]
Artur Cieślewicz (2021)	29/F	BNT162b2	First dose	1 day	No known past medical history	[31]
Ahmad Kantar (2021)	17/M	BNT162b2	First dose	12 hours	Allergic rhinitis	[34]
Talia Walter (2021)	43/M	BNT162b2	Second dose	10 hours	Atopy with seasonal rhinitis, eczema, and asthma	[42]
Parkash, Om (2021)	96/F	BNT162b2	First dose	Few days	Cholecystectomy	[32]
Ozaka, Sotaro (2021)	71/F	BNT162b2	First dose	2 days	Hypertension, hyperlipidemia, and cerebral infarction	[30]
Ammar A. Albokhari (2021)	15/F	BNT162b2	First dose	8 days	Diabetes mellitus type 1	[43]

Abbreviations and description: Female (F), Male (M).
[a] BNT162b2 (COVID-19 mRNA vaccine, BioNTech and Pfizer).
[b] The brand type was not reported in the study..
[c] 3–6 days.

Ethics approval

The study was approved by our local ethics committee.

Informed consent

Written informed consent was obtained from the patient for publication of this case report.

Author contribution statement

All authors listed have significantly contributed to the investigation, development and writing of this article.

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Data availability statement

Data will be made available on request.

Declaration of competing interest

The authors declare no conflict of interest.

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